

IoT and Real-Time Embedded Systems Lab

Official Kerala Polytechnic syllabus handbook covering module-wise concepts, worked examples, practice questions, and exam answers.

Course Code

5528

Credits

4

Periods

5/week

Course Details

Title: IoT and Real-Time Embedded Systems Lab

Department: Common

Revision: REV_IoT and Real-Time Embedded Systems Lab

Pattern: Theory / Practical

Complies with official SITTTR guidelines with Malayalam support notes and worked exercises.

Curriculum integrity

Official Source Declaration

This handbook's course identity is aligned with the Revision 2026 master subject index for **Course 5528 — IoT and Real-Time Embedded Systems Lab**. Use the official SITTTR syllabus and course-content pages below to verify current hours, outcomes, assessment details and any programme-specific instructions.

MODULE 1

Core Principles

Outcomes

Master foundational terminology and core definitions of IoT and Real-Time Embedded Systems Lab.

Study

Detailed analysis of core principles and theoretical foundations.

Malayalam Support Note: ഐഐഐഐ ഐഐഐഐഐഐഐഐ ഐഐഐഐഐഐഐഐ ഐഐഐഐഐഐഐഐ
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MODULE 2

Applied Methods

Outcomes

Apply standard calculation techniques and operational procedures.

Study

Numerical problem-solving techniques and practical workflows.

Malayalam Support Note: ഐഐഐഐഐഐഐ ഐഐഐഐഐഐഐഐഐഐഐഐഐഐഐഐഐഐഐഐഐ ഐഐഐഐഐഐഐഐ
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MODULE 3

Advanced Integration

Outcomes

Evaluate complex systems and optimize performance.

Study

Advanced system integration and troubleshooting methodologies.

Malayalam Support Note: ഐക്യരൂപം സൃഷ്ടിക്കുന്നതിനായി സാങ്കേതികവിദ്യകളെ ഉപയോഗിച്ച് സിസ്റ്റം ഇൻ്റഗ്രേഷൻ നടത്തേണ്ടതും സിസ്റ്റം പ്രവർത്തനം മെച്ചപ്പെടുത്തേണ്ടതും.

MODULE 4

Synthesis & Case Studies

Outcomes

Design comprehensive technical solutions and demonstrate mastery.

Study

Real-world case studies and industry project design.

Malayalam Support Note: സാങ്കേതികവിദ്യകളെ ഉപയോഗിച്ച് സിസ്റ്റം ഇൻ്റഗ്രേഷൻ നടത്തേണ്ടതും സിസ്റ്റം പ്രവർത്തനം മെച്ചപ്പെടുത്തേണ്ടതും.

Practice Model Question Paper

Time: 3 Hours | Max Marks: 75

Part A (2 marks each)

1. Define core concepts of IoT and Real-Time Embedded Systems Lab.
2. Explain fundamental principles in Module 1.
3. Discuss calculation methods in Module 2.
4. Describe advanced integration in Module 3.
5. Summarize case studies in Module 4.

Part B (5 marks each)

1. Detailed explanation of Module 1 concepts with examples.
2. Comprehensive analysis of Module 2 methods.
3. System integration and troubleshooting in Module 3.
4. Industry applications and design in Module 4.

Full Answer Key

Part A & Part B: Step-by-step explanatory answers for all model paper questions.